

Jinwoo Choi

CONTACT INFORMATION

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RESEARCH INTERESTS

**Architecture and system optimization for neural networks /
Computer architecture, system software, and hardware acceleration**
On-device inference, Scheduling, Neural Processing Unit (NPU), Performance Modeling

EDUCATION

Yonsei University, Seoul, Korea,
Ph.D. Candidate, Computer Science **Mar. 2020 - August. 2025**
• Advisor: Prof. Youngsok Kim
B.S., Computer Science **Mar. 2015 - Feb. 2020**

RESEARCH EXPERIENCE

Post Doctoral Researcher **Sep. 2025 - Present**
Visual Intelligence Group (VIG) (Center for AI, Korea Institute of Science and Technology (KIST))
Graduate Researcher **Mar. 2020 - Aug. 2025**
Advisor: Prof. Youngsok Kim (High Performance Computing Platforms Lab, Yonsei University)

PUBLICATIONS

Chaemin Lim, Suhyun Lee, **Jinwoo Choi**, Joonsung Kim, Jinho Lee, and Youngsok Kim, “FaSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables”, In *Proc. 42nd IEEE International Conference on Data Engineering (ICDE)*, May. 2026.

Chaemin Lim, Suhyun Lee, **Jinwoo Choi**, Joonsung Kim, Jinho Lee, and Youngsok Kim, “DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP”, In *Proc. 34th International Conference on Parallel Architectures and Compilation Techniques (PACT)*, Nov. 2025.

Suhyun Lee, Chaemin Lim, **Jinwoo Choi**, Heelim Choi, Chan Lee, Yongjun Park, Kwanghyun Park, Hanjun Kim, and Youngsok Kim, “SPID-Join: A Skew-resistant Processing-in-DIMM Join Algorithm Exploiting the Bank- and Rank-level Parallelisms of DIMMs”, In *Proc. 2025 ACM International Conference on Management of Data (SIGMOD)*, June 2025.

Jinwoo Choi¹, Yeonan Ha¹, Hanna Cha, Seil Lee, Sungchul Lee, Jounghoo Lee, Shin-haeng Kang, Bongjun Kim, Hanwoong Jung, Hanjun Kim and Youngsok Kim, “MPC-Wrapper: Fully Harnessing the Potential of Samsung Aquabolt-XL HBM2-PIM on FPGAs”, In *Proc. 32nd IEEE International Symposium On Field-Programmable Custom Computing Machines (FCCM)*, May. 2024.

¹Co-first authors

Jinwoo Choi, Yeonan Ha, Jounghoo Lee, Sangsu Lee, Jinho Lee, Hanhwi Jang, and Youngsok Kim, “Enabling Fine-Grained Spatial Multitasking on Systolic-Array NPUs Using Dataflow Mirroring”, *IEEE Transactions on Computers (TC)*, Aug. 2023

Chaemin Lim, Suhyun Lee, **Jinwoo Choi**, Jounghoo Lee, Seongyeon Park, Hanjun Kim, Jinho Lee, and Youngsok Kim, “Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs”, In *Proc. 2023 ACM International Conference on Management of Data (SIGMOD)*, June 2023.

Jinwoo Choi¹, Jaeyeon Kim¹, Chaemin Lim¹, Suhyun Lee, Jinho Lee, Dokyung Song, and Youngsok Kim, “GuardiaNN: Fast and Secure On-Device Inference in TrustZone Using Embedded SRAM and Cryptographic Hardware”, In *Proc. 23rd ACM/IFIP International Middleware Conference (Middleware)*, Nov. 2022.

¹Co-first authors

Jounghoo Lee¹, **Jinwoo Choi**¹, Jaeyeon Kim, Jinho Lee, and Youngsok Kim, “Dataflow Mirroring: Architectural Support for Highly Efficient Fine-Grained Spatial Multitasking on Systolic-Array NPU’s”, In *58th ACM/IEEE Design Automation Conference (DAC)*, Dec. 2021.

¹Co-first authors

PATENTS

[**Registration: JP 2025025217**] Youngsok Kim, **Jinwoo Choi**, Chaemin Lim, Suhyun Lee, Hanjun Kim, ”Skew Resistance Processing in DIMM Device and Processing Method”, *Japan Patent*

[**Registration: JP 2021168073**] Youngsok Kim, **Jinwoo Choi**, Jounghoo Lee, ”Method and Apparatus for Performing Deep Learning Operations”, *Japan Patent*

[**Registration: KR 1020200144563**] Youngsok Kim, **Jinwoo Choi**, Jounghoo Lee, ”Method and Apparatus for Performing Deep Learning Operations”, *Korea Patent*

[**Registration: JP 2023129058**] Youngsok Kim, **Jinwoo Choi**, Chaemin Lim, Suhyun Lee, Dokyung Song, Jinho Lee, ”Artificial Intelligence Device Based on Trust Environment”, *Japan Patent*

[**Registration: KR 1020230058602**] Youngsok Kim, **Jinwoo Choi**, Chaemin Lim, Suhyun Lee, Dokyung Song, Jinho Lee, ”Artificial Intelligence Device Based on Trust Environment”, *Korea Patent*

PROJECTS

Exploring Quality-Aware NN Scheduler

Mar. 2024 - Aug. 2025

- LG Electronics

Designing a PIM-Based NPU Architecture and FPGA Prototyping

Mar. 2023 - Feb. 2024

- Samsung Advanced Institute of Technology (SAIT)

Sparsity-Aware Spatial Multitasking NPU Architecture and Scheduling

Jun. 2022 - May 2023

- National Research Foundation of Korea (NRF)

Developing Software Platform for Programming of PIM

Jun. 2021 - Dec. 2023

- Institute for Information & communications Technology Promotion (IITP)

Smart, Attack-resistant IoT Networks

Dec. 2020 - Nov. 2023

- Korea Institute for Advancement of Technology (KIAT)

Spatial Multitasking on Systolic Array Neural Processing Units

Mar. 2020 - Feb. 2021

- Samsung Advanced Institute of Technology (SAIT)

AWARDS AND HONORS

KIST Postdoctoral Research Fellowship

May. 2026

- Korea Institute of Science and Technology

Best Paper - Yonsei University Innovation Paper Award

Jul. 2023

- Yonsei University Graduate School

Doctor Fellowship from NRF of Korea

May 2022

- National Research Foundation (NRF) of Korea

SKILLS

Programming Languages

- C, C++, Python

Tools

- Deep Learning Frameworks (TensorFlow, TensorFlow Lite, PyTorch, TensorRT, MACE)
- Simulators (SCALE-SIM, DRAMSim3, OpenRoad Flow, Accelergy, CACTI-7)

Languages

- Korean(native), English